

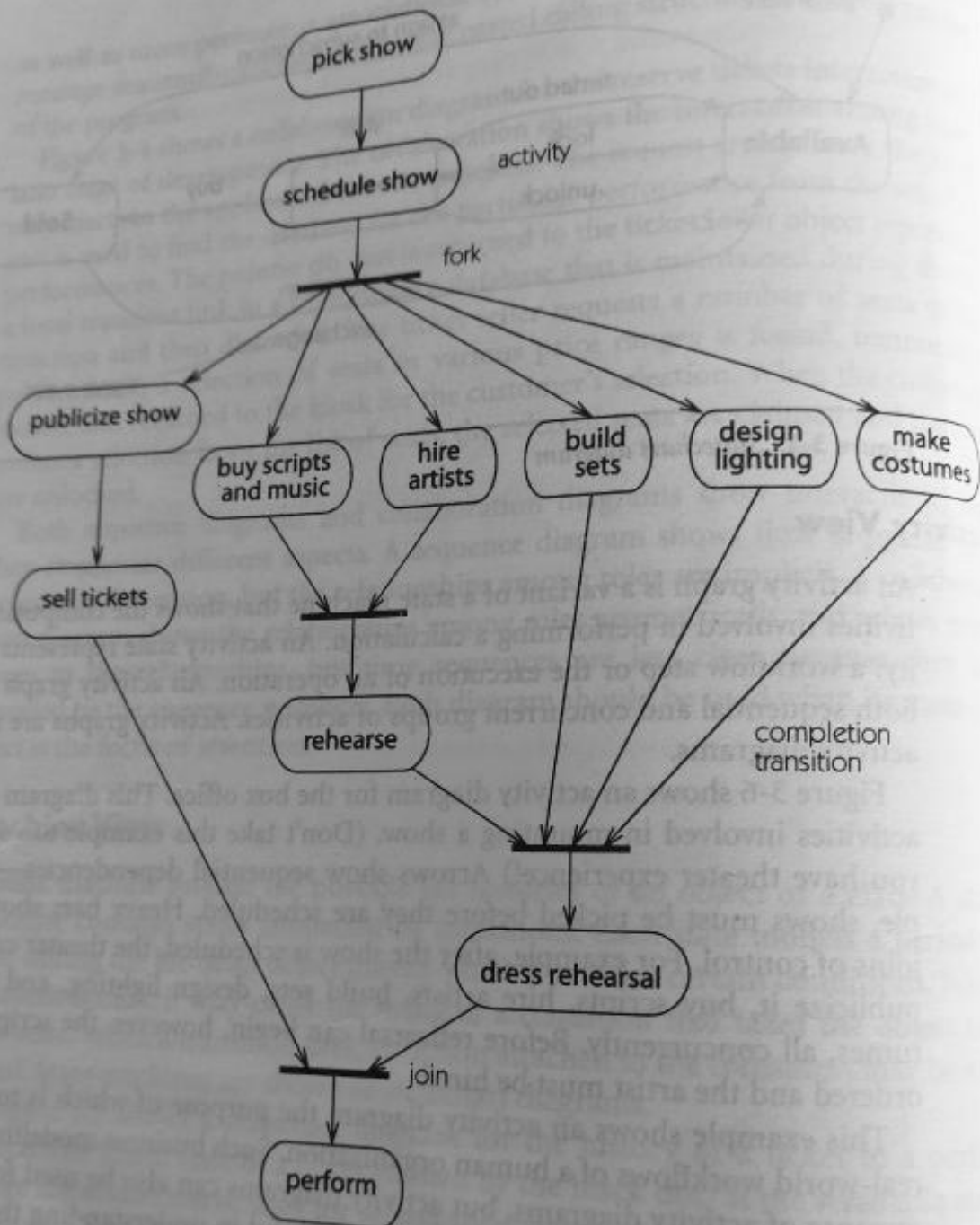


Figure 3-6. Activity diagram

Physical Views

The previous views model the concepts in the application from a logical viewpoint. The physical views model the implementation structure of the application itself, such as its organization into components and its deployment onto run-time nodes. These views provide an opportunity to map classes onto implementation

components and nodes. There are two physical views: the implementation view and the deployment view.

The implementation view models the components in a system—that is, the software units from which the application is constructed—as well as the dependencies among components so that the impact of a proposed change can be assessed. It also models the assignment of classes and other model elements to components.

The implementation view is displayed on component diagrams. Figure 3-7 shows a component diagram for the box office system. There are three user

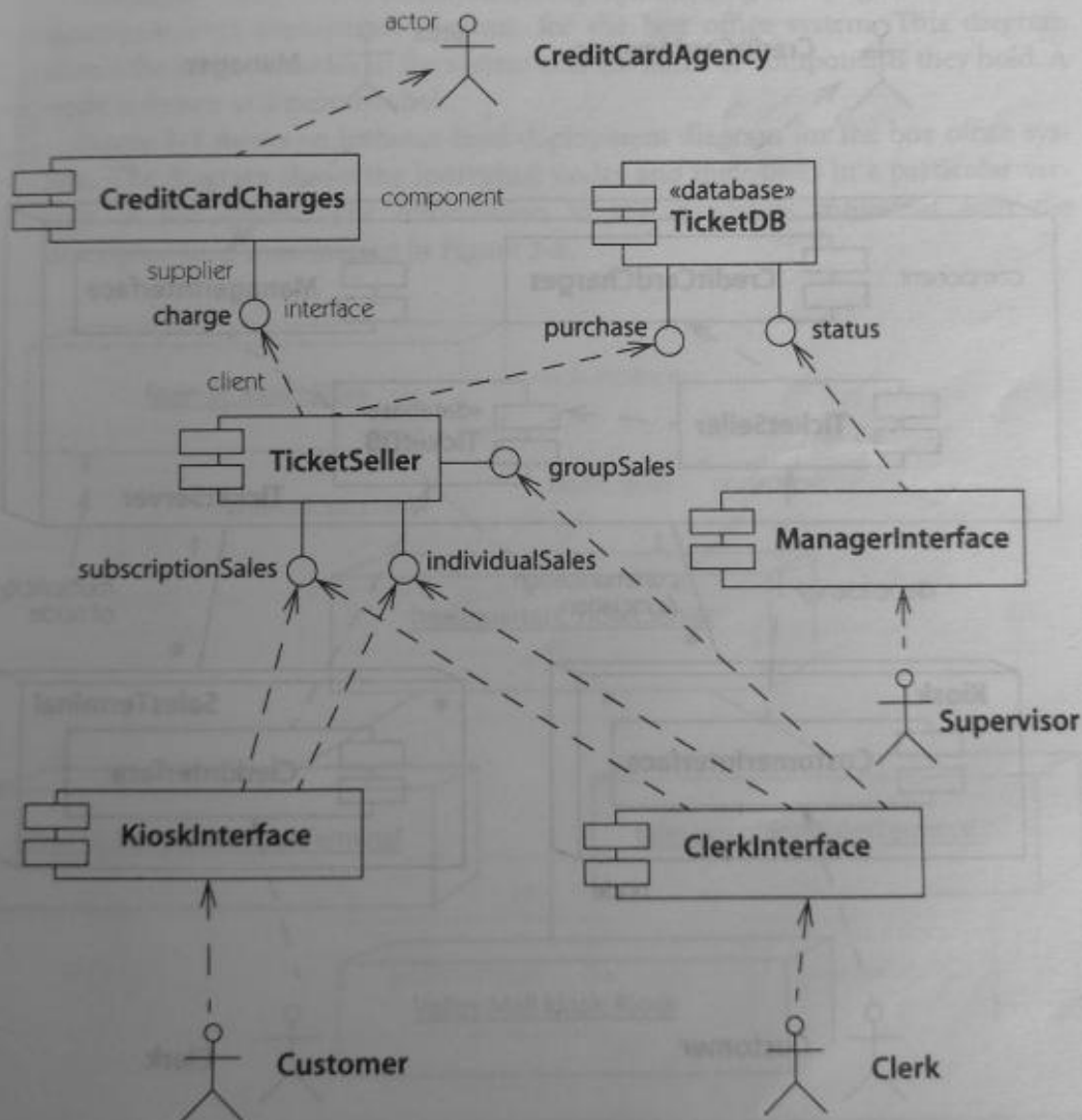


Figure 3-7. Component diagram

interfaces: one each for customers using a kiosk, clerks using the on-line reservation system, and supervisors making queries about ticket sales. There is a ticket seller component that sequentializes requests from both kiosks and clerks; a component that processes credit card charges; and the database containing the ticket information. The component diagram shows the kinds of components in the system; a particular configuration of the application may have more than one copy of a component.

A small circle with a name is an interface—a coherent set of services. A solid line from a component to an interface indicates that the component provides the service.

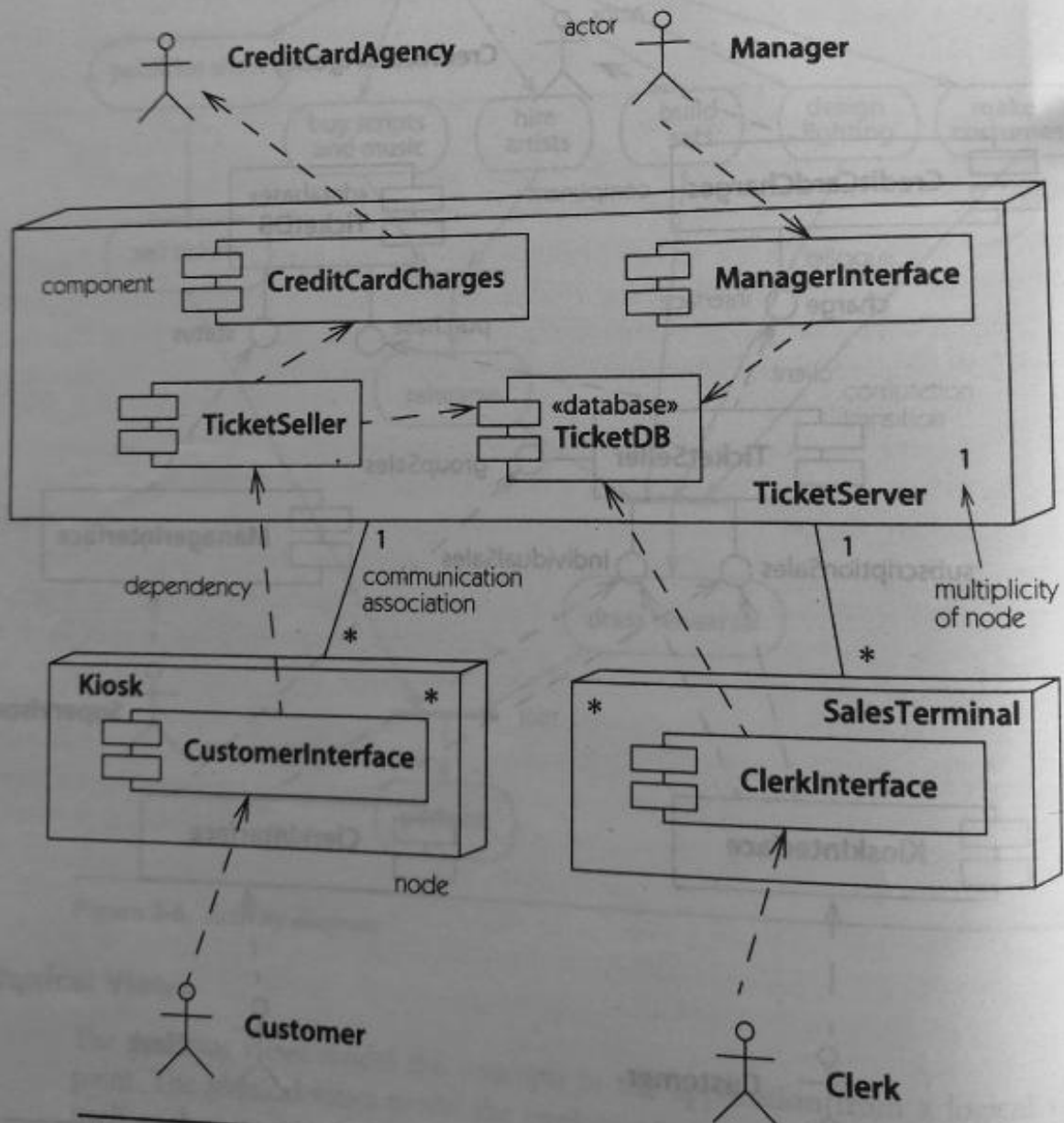


Figure 3-8. Deployment diagram (descriptor level)

vices listed in the interface. A dashed arrow from a component to an interface indicates that the component requires the services provided by the interface. For example, subscription sales and group sales are both provided by the ticket seller component; subscription sales are accessible from both kiosks and clerks, but group sales are only accessible from a clerk.

The deployment view represents the arrangement of run-time component instances on node instances. A node is a run-time resource, such as a computer, device, or memory. This view permits the consequences of distribution and resource allocation to be assessed.

The deployment view is displayed on deployment diagrams. Figure 3-8 shows a descriptor-level deployment diagram for the box office system. This diagram shows the kinds of nodes in the system and the kinds of components they hold. A node is shown as a cube symbol.

Figure 3-9 shows an instance-level deployment diagram for the box office system. The diagram shows the individual nodes and their links in a particular version of the system. The information in this model is consistent with the descriptor-level information in Figure 3-8.

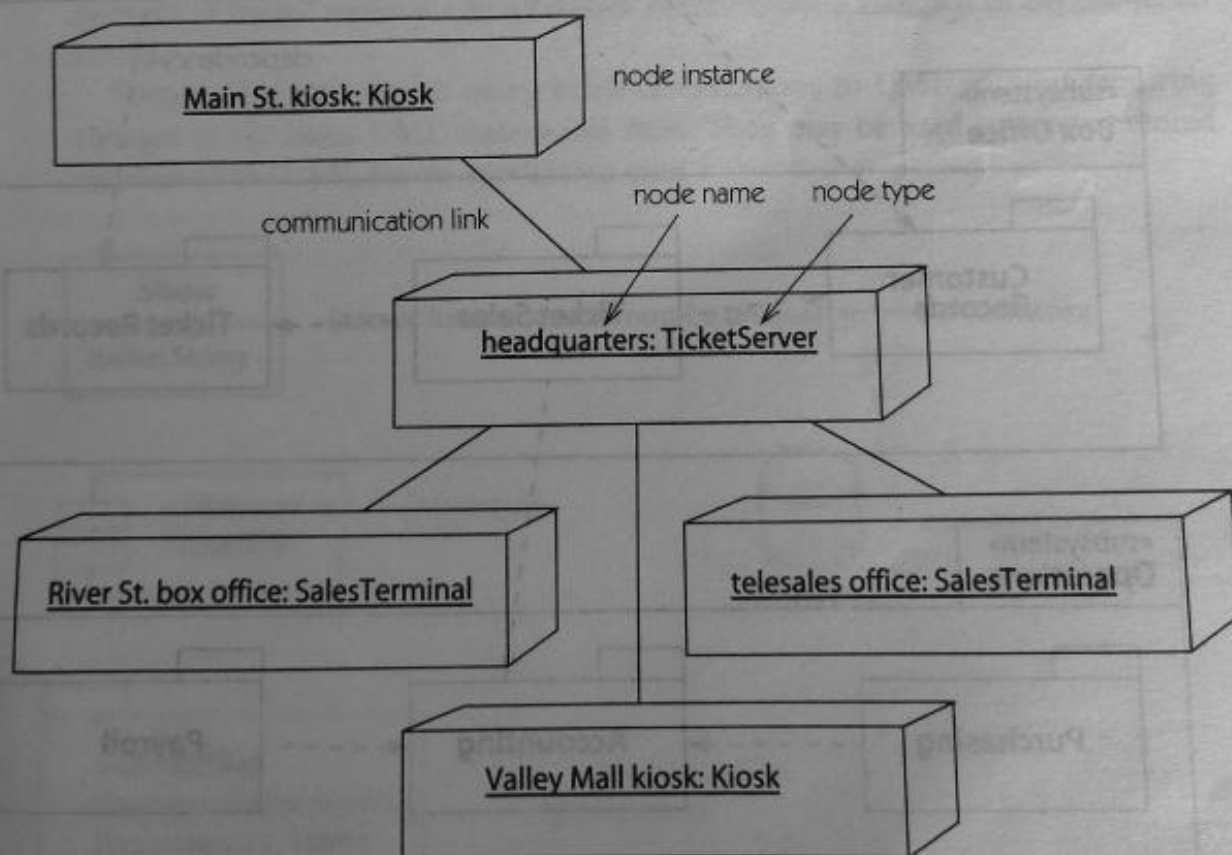


Figure 3-9. Deployment diagram (instance level)

Model Management View

The model management view models the organization of the model itself. A model comprises a set of packages that hold model elements, such as classes, state machines, and use cases. Packages may contain other packages: therefore, a model designates a root package that indirectly contains all the contents of the model. Packages are units for manipulating the contents of a model, as well as units for access control and configuration control. Every model element is owned by one package or one other element.

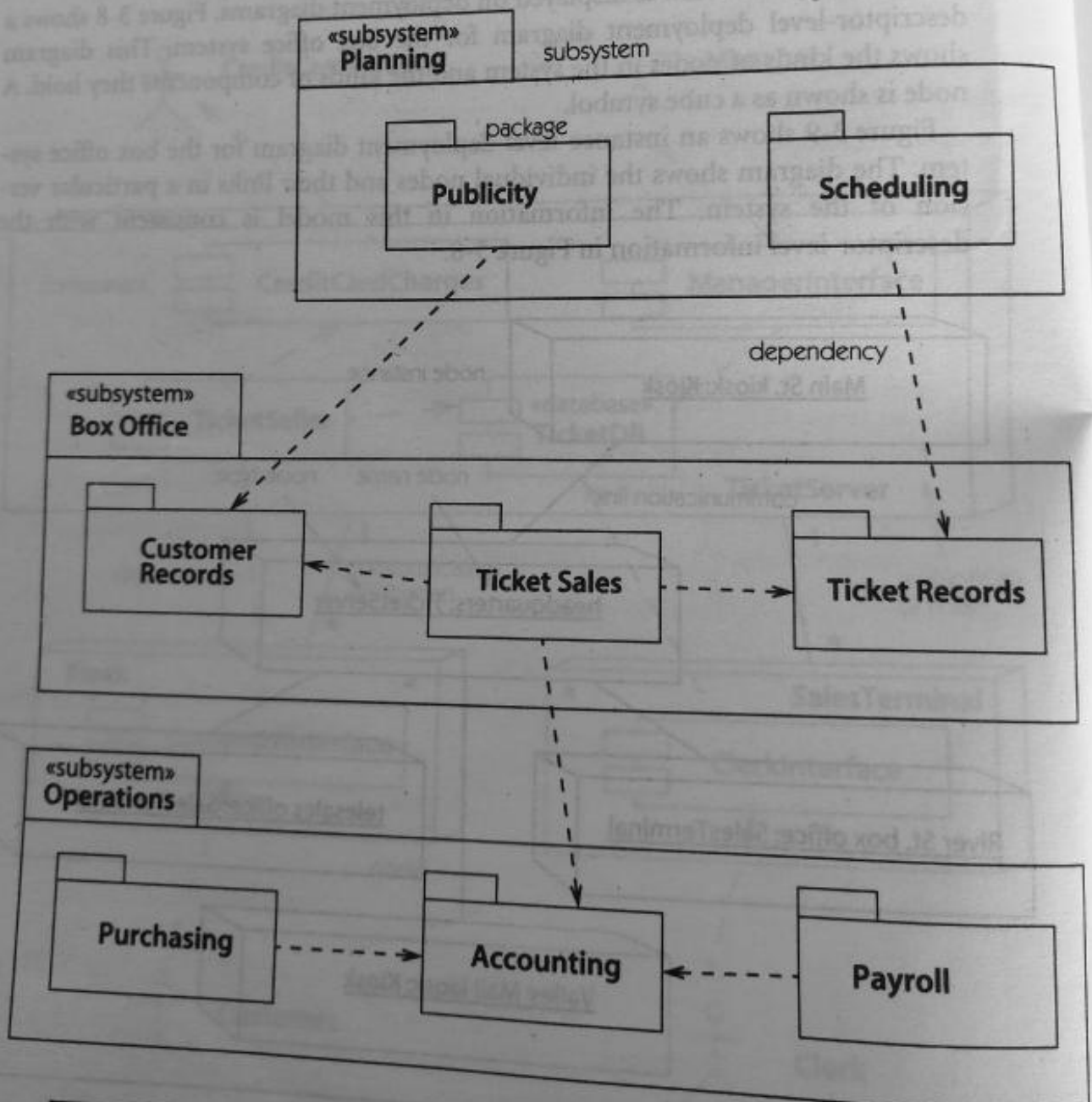


Figure 3-10. Packages

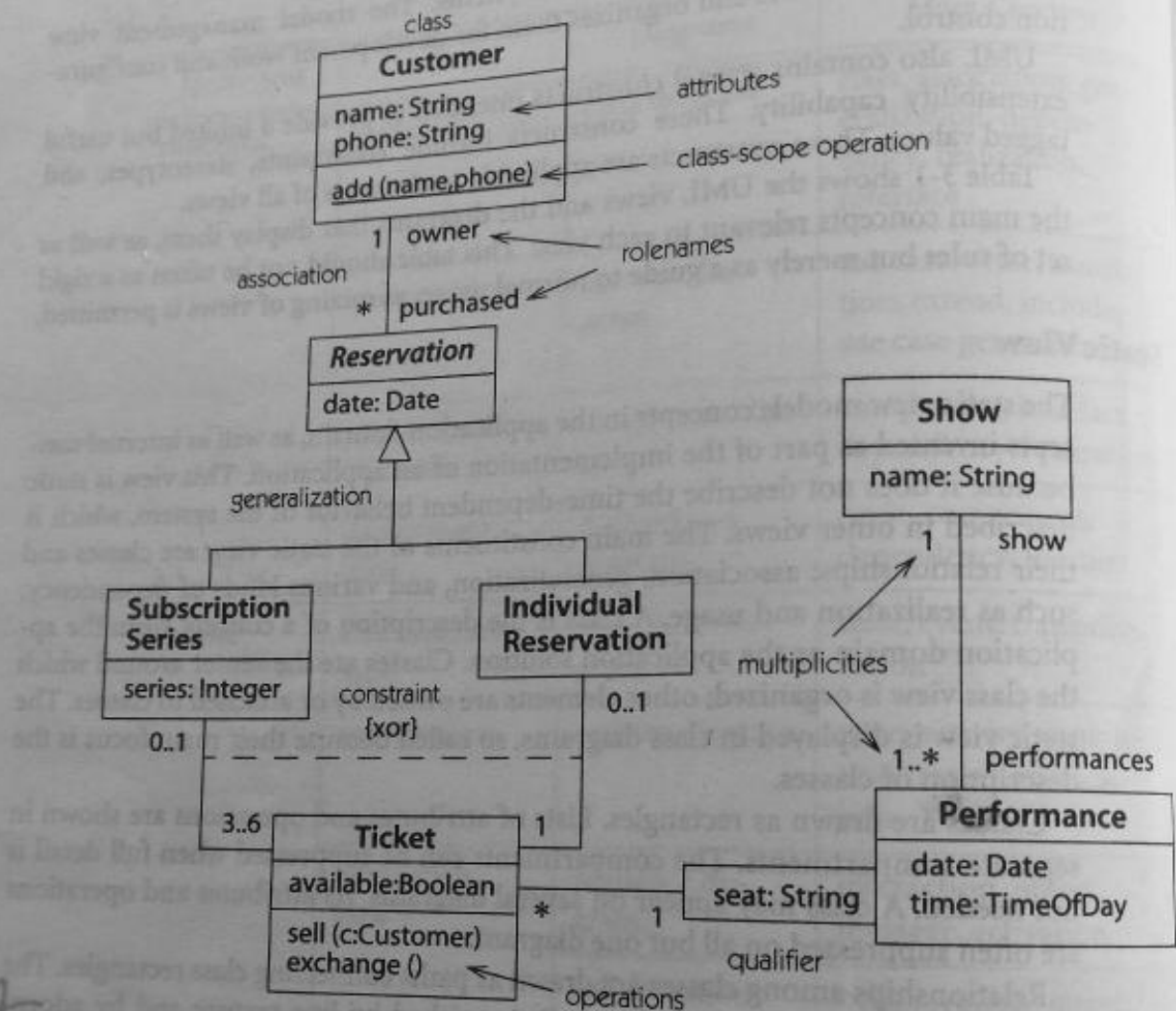


Figure 3-1. Class diagram

Use Case View

The use case view models the functionality of the system as perceived by outside users, called actors. A use case is a coherent unit of functionality expressed as a transaction among actors and the system. The purpose of the use case view is to list the actors and use cases and show which actors participate in each use case.

Figure 3-2 shows a use case diagram for the box office example. Actors include the clerk, supervisor, and kiosk. The kiosk is another system that accepts orders from a customer. The customer is not an actor in the box office application because the customer is not directly connected to the application. Use cases include buying tickets through the kiosk or the clerk, buying subscriptions (only through the clerk), and surveying total sales (at the request of the supervisor). Buying tick-

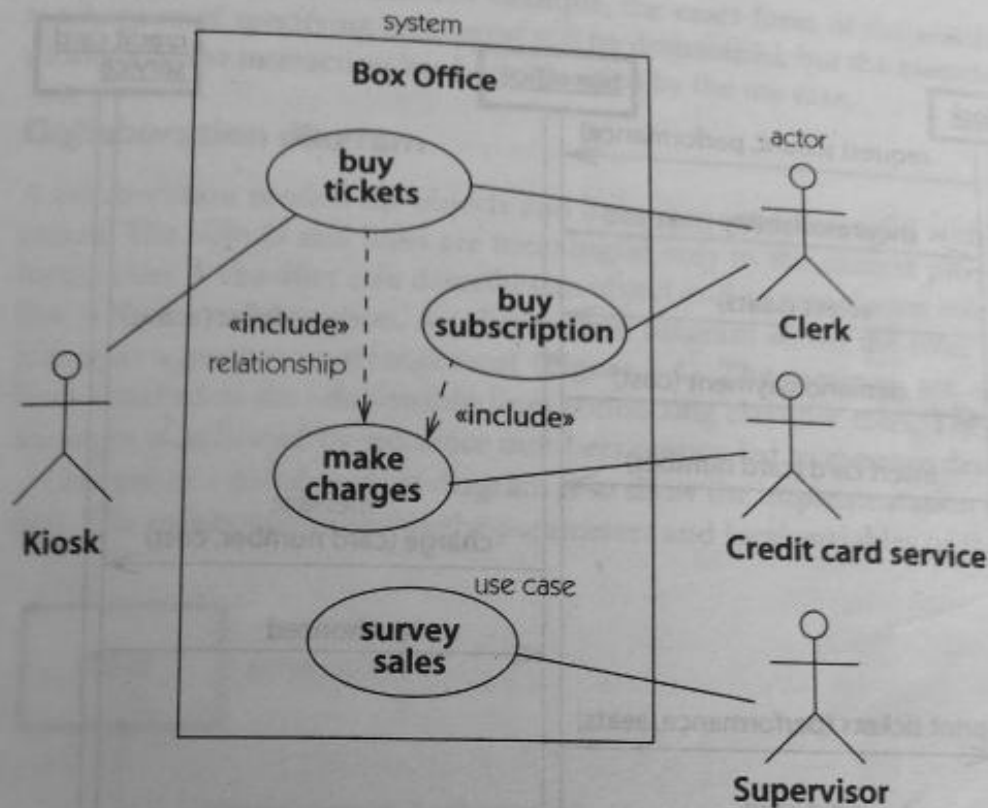


Figure 3-2. Use case diagram

ets and buying subscriptions include a common fragment—that is, making charges to the credit card service. (A complete description of a box office system would involve a number of other use cases, such as exchanging tickets and checking availability.)

Use cases can also be described at various levels of detail. They can be factored and described in terms of other, simpler use cases. A use case is implemented as a collaboration in the interaction view.

Interaction View

The interaction view describes sequences of message exchanges among roles that implement behavior of a system. A classifier role is the description of an object that plays a particular part within an interaction, as distinguished from other objects of the same class. This view provides a holistic view of behavior in a system—that is, it shows the flow of control across many objects. The interaction view is displayed in two diagrams focused on different aspects: sequence diagrams and collaboration diagrams.

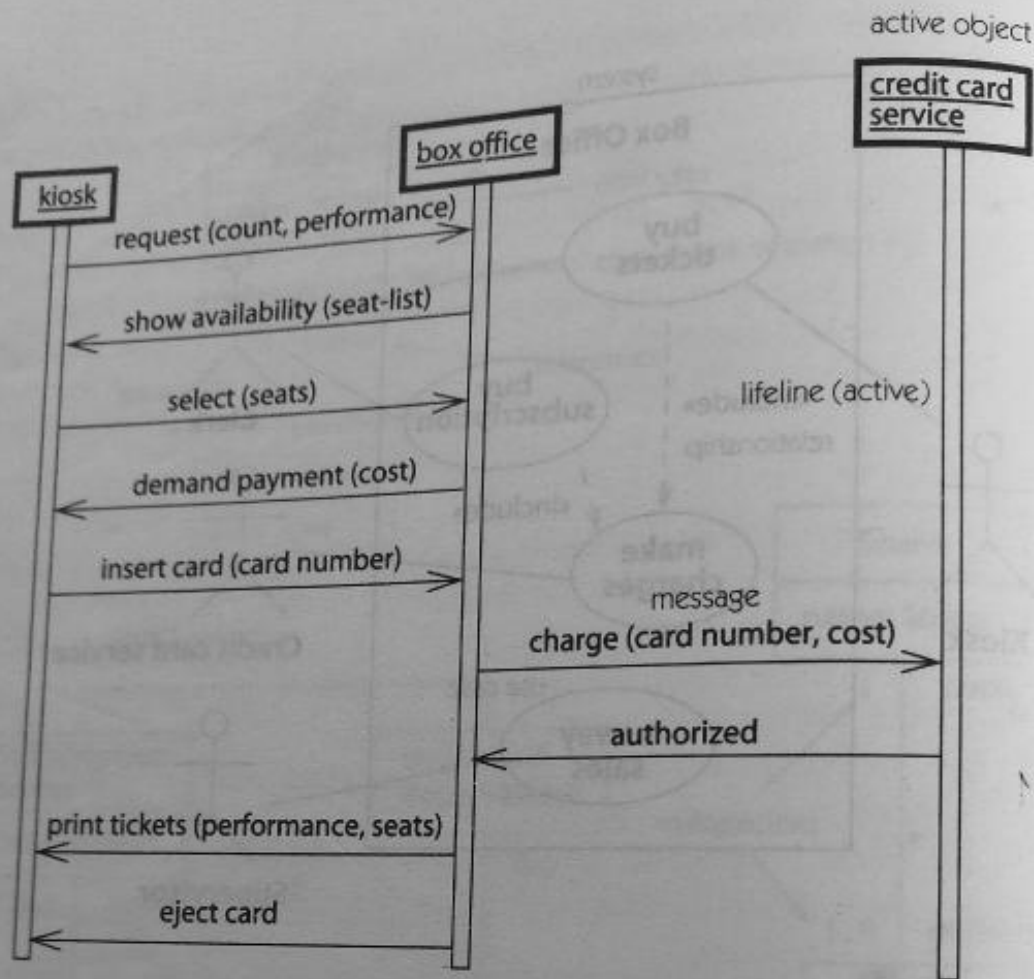


Figure 3-3. Sequence diagram

Sequence diagram

A sequence diagram shows a set of messages arranged in time sequence. Each classifier role is shown as a lifeline—that is, a vertical line that represents the role over time through the entire interaction. Messages are shown as arrows between lifelines. A sequence diagram can show a scenario—that is, an individual history of a transaction.

One use of a sequence diagram is to show the behavior sequence of a use case. When the behavior is implemented, each message on a sequence diagram corresponds to an operation on a class or an event trigger on a transition in a state machine.

Figure 3-3 shows a sequence diagram for the buy tickets use case. This use case is initiated by the customer at the kiosk communicating with the box office. The steps for the make charges use case are included within the sequence, which involves communication with both the kiosk and the credit card service. This sequence diagram is at an early stage of development and does not show the full